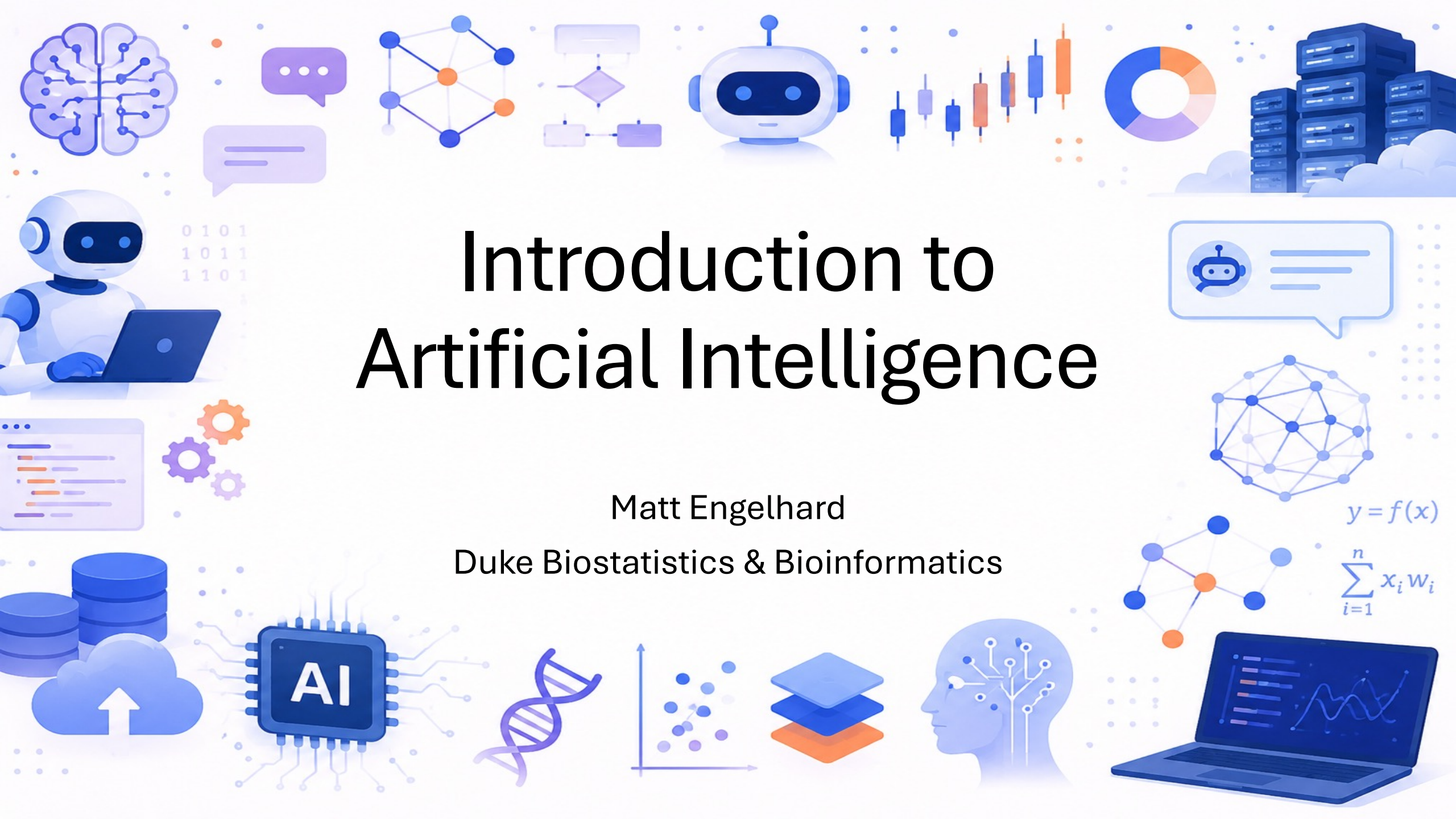




Introduction to Artificial Intelligence

Matt Engelhard

Duke Biostatistics & Bioinformatics

$$y = f(x)$$

$$\sum_{i=1}^n x_i w_i$$

About Matt



Duke Undergraduate

MD/PhD at UVA



Duke B&B Faculty Since 2021

AI for enhanced neurodevelopmental support

Problem:

Inconsistent recognition and referral



**Relevant
info is
scattered
across
the EHR**

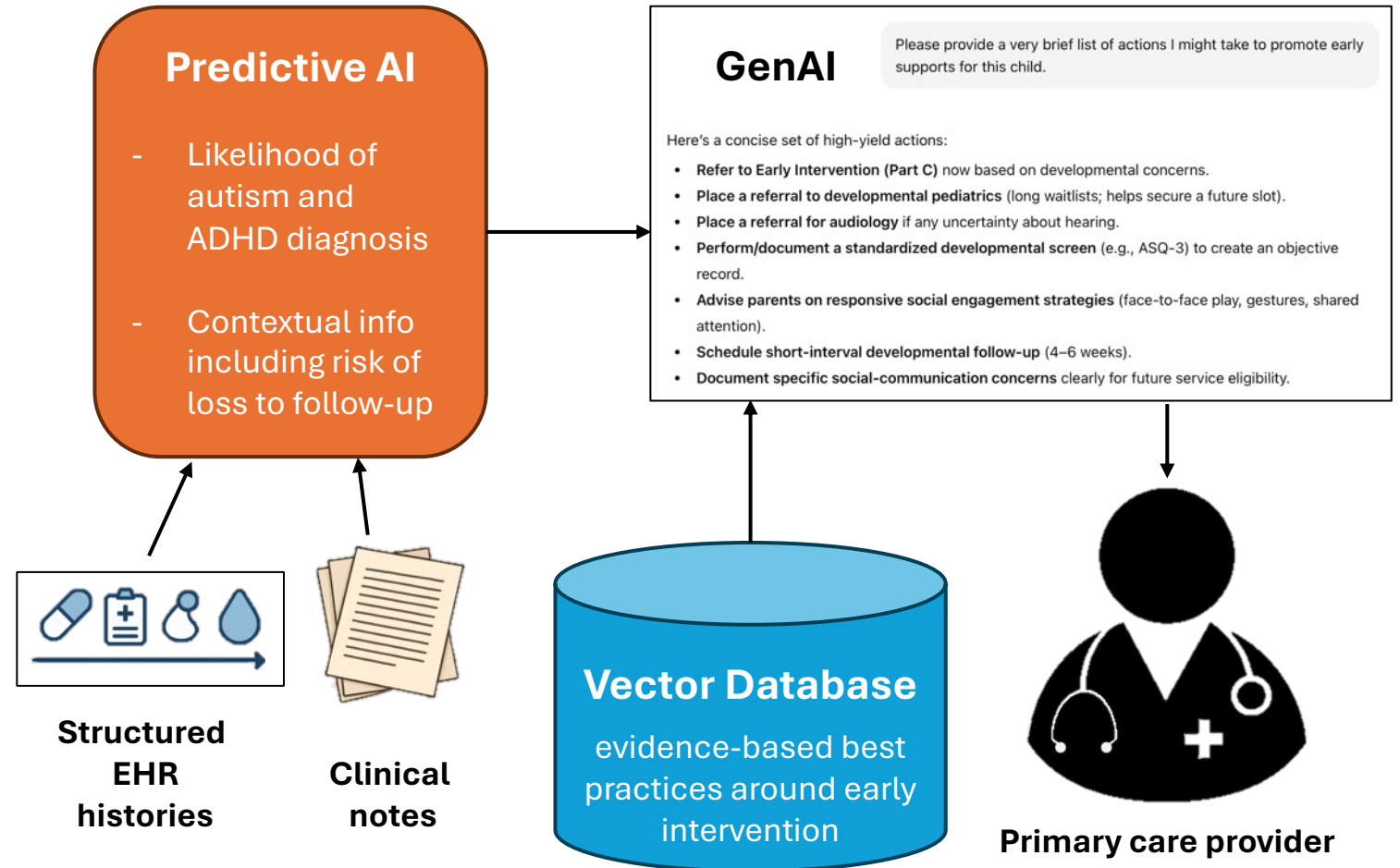
**Providers
need tailored
guidance
supporting
early action
and
intervention**

“Clinicians wanted a breakdown of **specific EHR data** that informed autism likelihood predictions, along with an explanation of how the AI-based system worked. Clinicians also wanted **actionable guidance** on next steps following a positive screen”

Emovon, Driggers-Jones, Engelhard, Maslow, Dawson, Goldstein, and Franz. Understanding End-user Contexts and Identifying Design Preferences of an AI-based Clinical Decision Support Tool for Early Autism Detection. Submitted to JAMIA Open.

Solution:

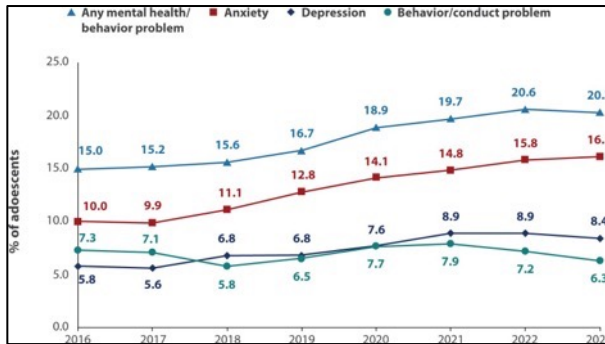
AI-based recommendations supporting personalized early actions



AI to improve rural adolescent mental health

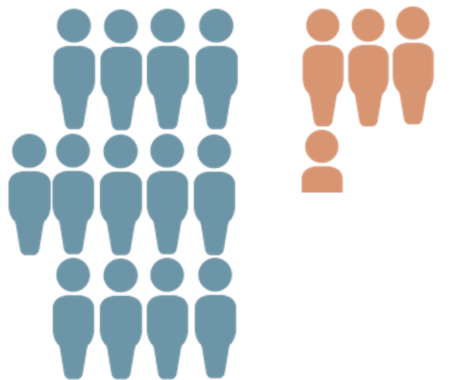
Problem:

Mental health crisis in rural youth



Increasing mental health diagnoses among children ages 12-17

Profound shortage of mental health providers in rural areas

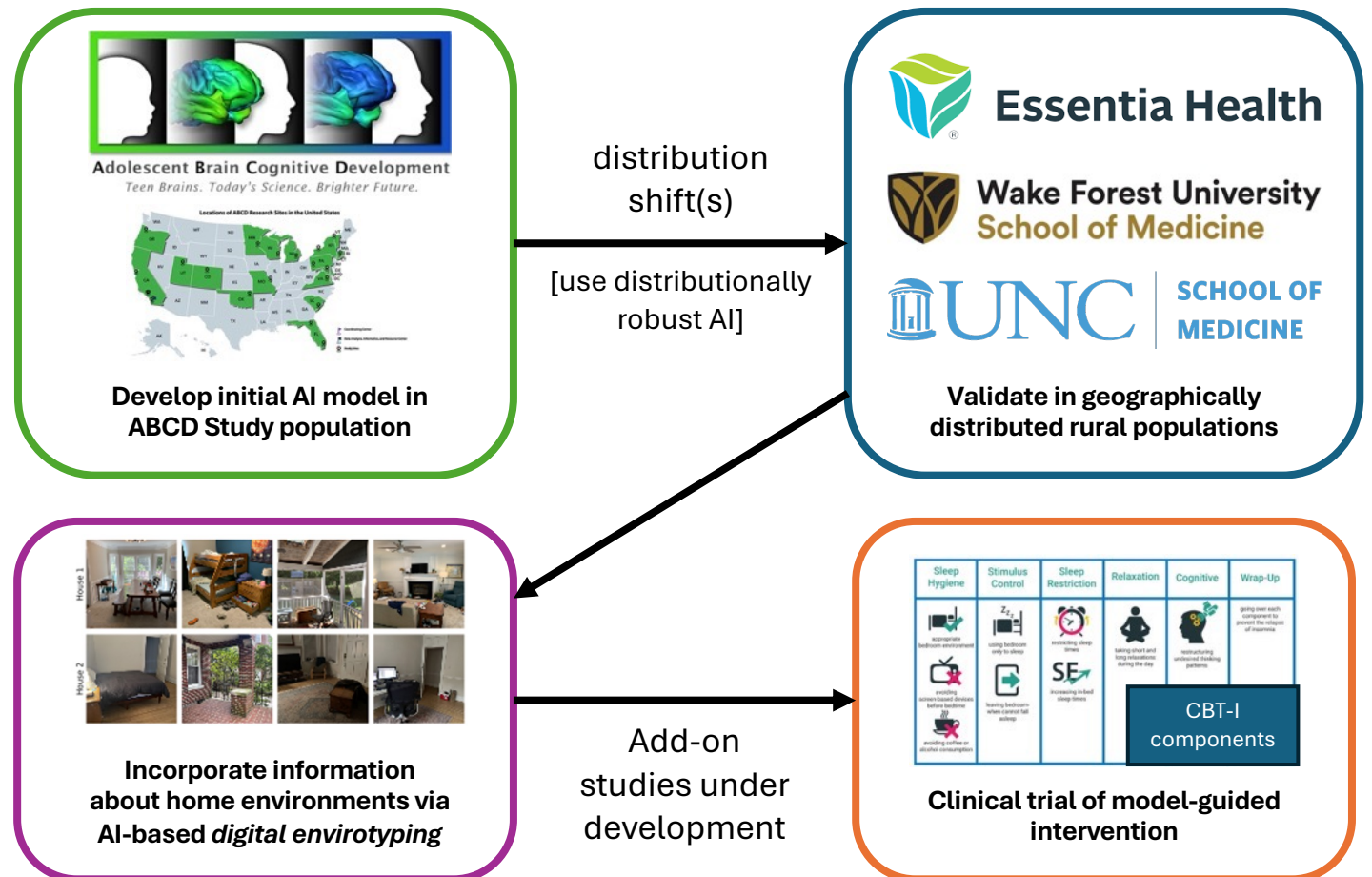


Urban:
13 per 100k

Rural:
3.5 per 100k

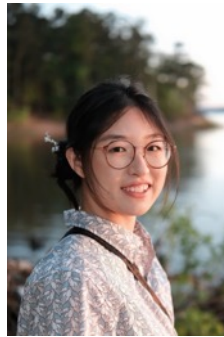
Solution:

Mechanistic AI model that identifies high-risk youth and suggests possible intervention target(s)

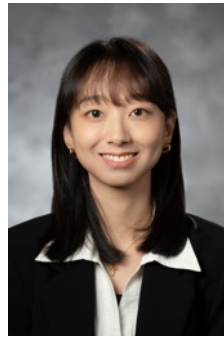


Engelhard Lab: patient-centered, AI-based CDS

Current
PhD
students



Qin
Weng



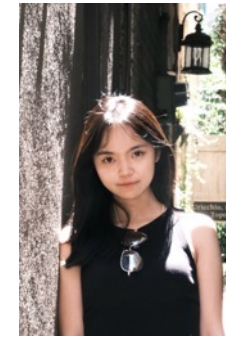
Liwen
Sun



Fengnan
Li



Yuqi
Li

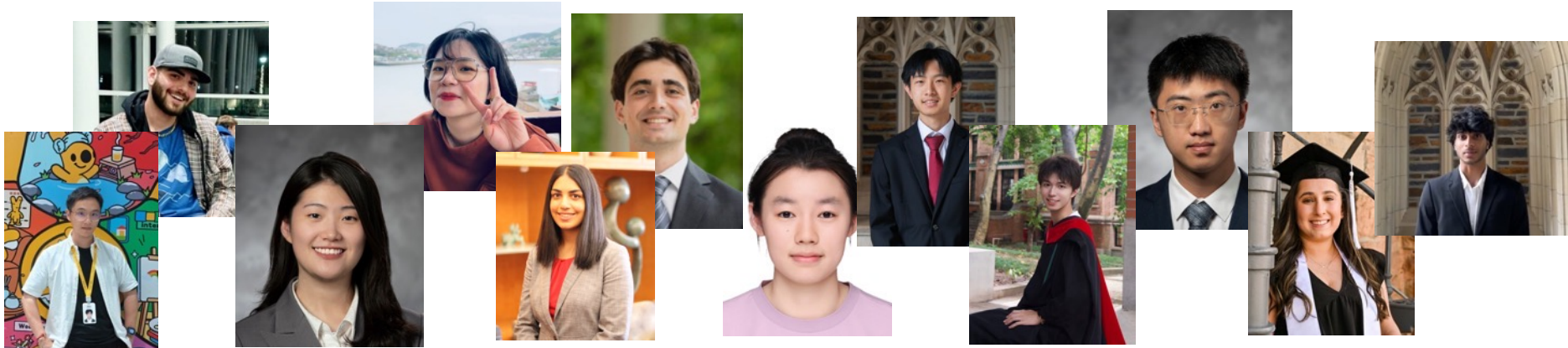


Pengxi
Liu



Yuankang
Zhao

Current
MS/MB &
medical
students



About Andrea



MMCI Health Summit in DC, 2021



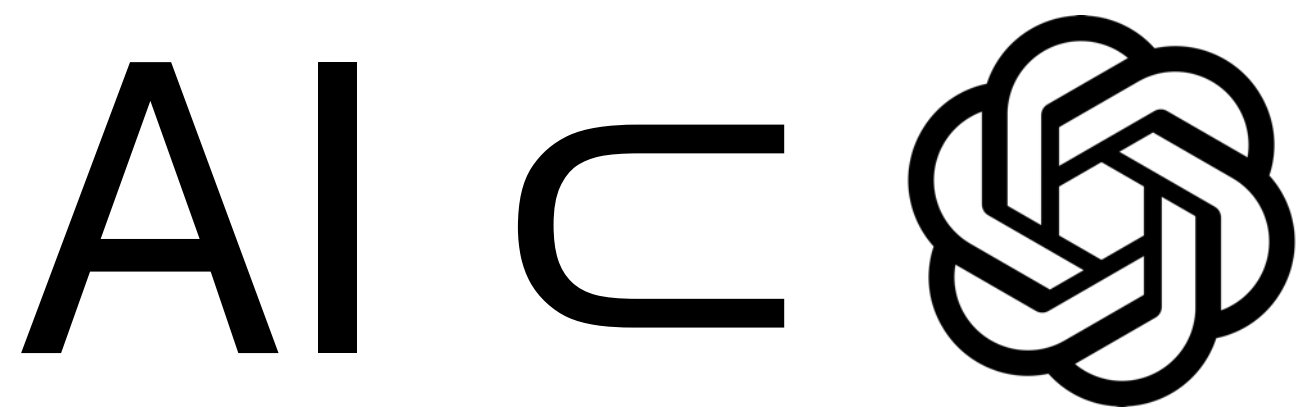
MMCI Class of 2021

About You!

- Your name
- Your most recent position/training
(one line)
- Are you skeptical or hopeful on impact of AI?
(1 = skeptical to 10 = hopeful)
- Why?
(very briefly, e.g. *it will make us dumber*, or *it will cure all disease*)

What is AI?

AI $\neq$ 

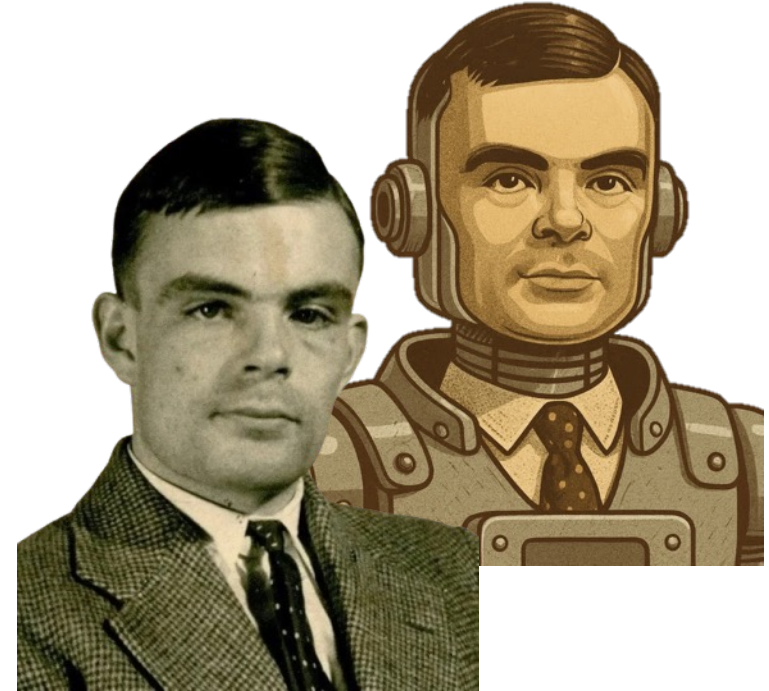


What is AI?

A scientific field concerned with developing **intelligent machines**.

What is **intelligence**?

- learn from experience
- solve problems
- make decisions



Turing or TuringBot?

“AI” through the (recent) years



2018: AI = computer vision

“AI” through the (recent) years



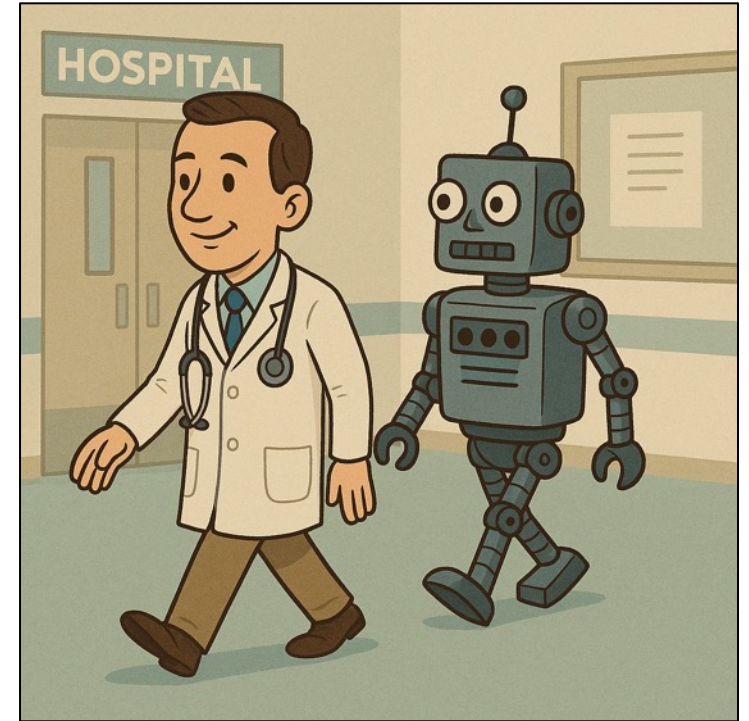
Please provide a very brief, bulleted summary of evidence supporting the effectiveness of the Early Start Denver Model for early autism intervention.

Here's a concise, evidence-focused, bulleted summary:

- **Improved cognitive and language outcomes:** Multiple randomized controlled trials (RCTs) show that children receiving the Early Start Denver Model (ESDM) demonstrate significantly greater gains in IQ,

2024: AI = LLMs/chatbots

“AI” through the (recent) years



2029: AI = agents and RL



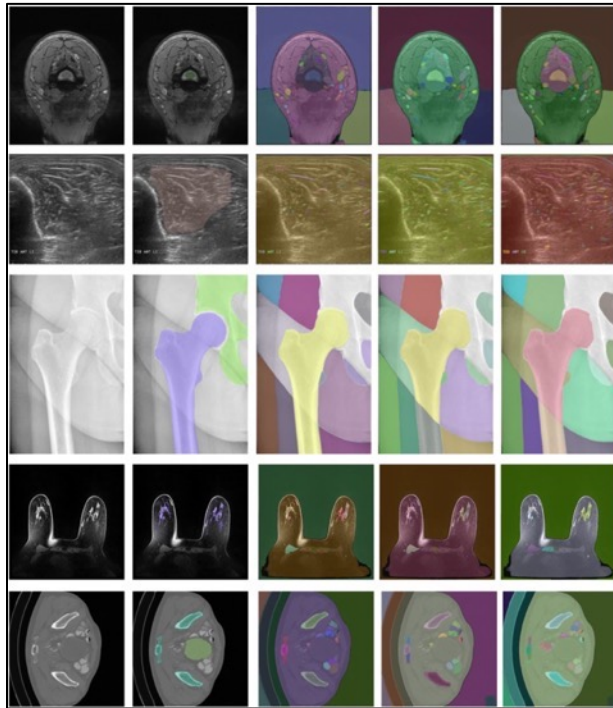
Please provide a very brief, bulleted summary of evidence supporting the effectiveness of the Early Start Denver Model for early autism intervention.

Here's a concise, evidence-focused, bulleted summary:

- **Improved cognitive and language outcomes:** Multiple randomized controlled trials (RCTs) show that children receiving the Early Start Denver Model (ESDM) demonstrate significantly greater gains in IQ,

Branches of AI with high clinical impact

Computer vision



Mazurowski, Maciej A., et al. "Segment anything model for medical image analysis: an experimental study." *Medical Image Analysis* 89 (2023): 102918.

Natural language processing

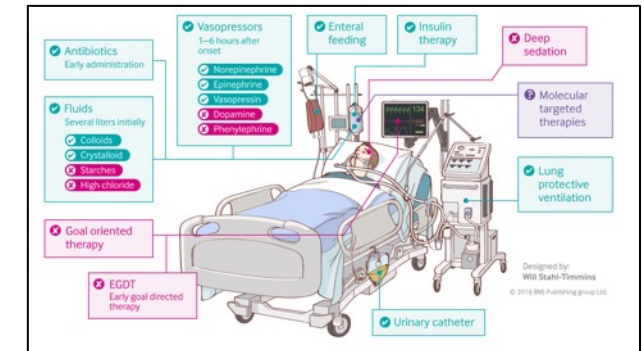


Please provide a very brief, bulleted summary of evidence supporting the effectiveness of the Early Start Denver Model for early autism intervention.

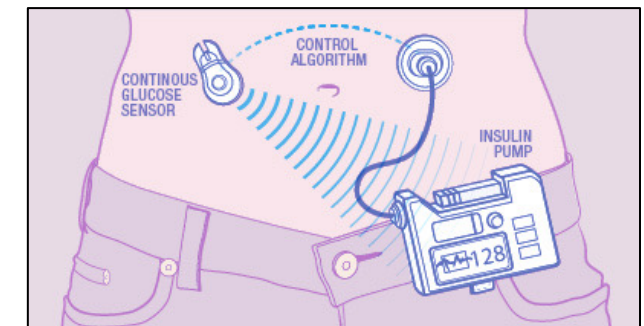
Here's a concise, evidence-focused, bulleted summary:

- **Improved cognitive and language outcomes:** Multiple randomized controlled trials (RCTs) show that children receiving the Early Start Denver Model (ESDM) demonstrate significantly greater gains in IQ, adaptive behavior, and expressive/receptive language compared with control groups receiving standard community services.
- **Enhanced adaptive functioning:** Studies report improvements in social communication, daily living skills, and overall adaptive behaviors, sustained beyond treatment.
- **Reduced autism symptom severity:** Follow-up analyses indicate modest but significant reductions in ASD symptom severity relative to baseline and comparison groups.
- **Neural evidence of typical processing:** EEG studies suggest that children treated with ESDM show more typical patterns of brain activity in response to social stimuli, indicating neural-level changes.
- **Generalization to community settings:** Effectiveness has been demonstrated in both controlled university-based settings and community-based early intervention programs, supporting scalability.
- **Durable effects:** Longitudinal follow-ups show persistence of developmental gains several years post-intervention.

Reinforcement learning

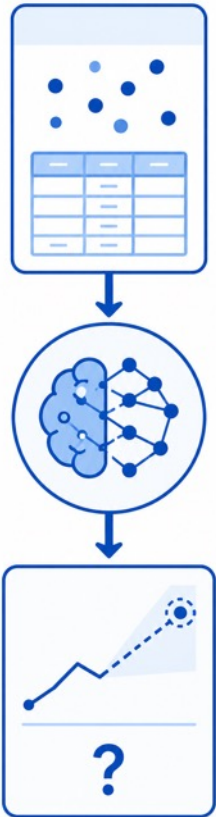


RL-based sepsis management



closed-loop BG control

Predictive AI versus Generative AI (GenAI)



Predictive AI

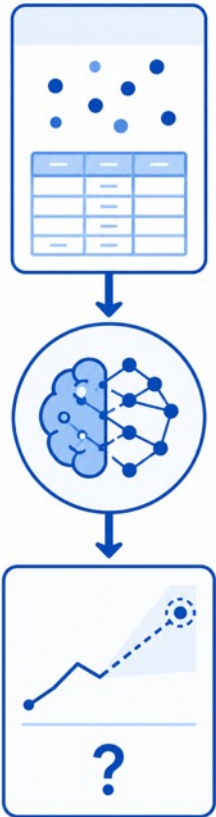
Tasks

- Regression
- Classification
- Risk prediction

Types of data

- Tabular
- Images
- Text
- Biomedical signals
- Multimodal

Predictive AI versus Generative AI (GenAI)



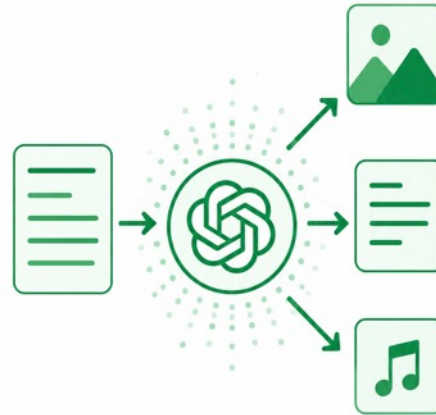
Predictive AI

Tasks

- Regression
- Classification
- Risk prediction

Types of data

- Tabular
- Images
- Text
- Biomedical signals
- Multimodal



Generative AI

Tasks

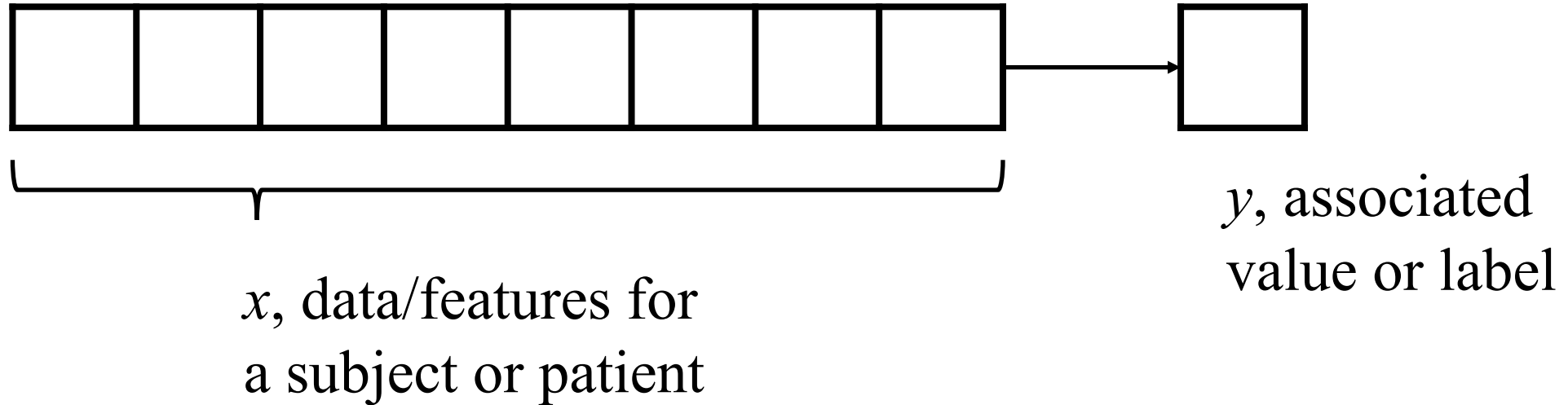
- “Data generation”
- Less well-defined
- Can perform pred tasks

Types of data

- Tabular
- Images
- **Text**
- Biomedical signals
- **Multimodal**

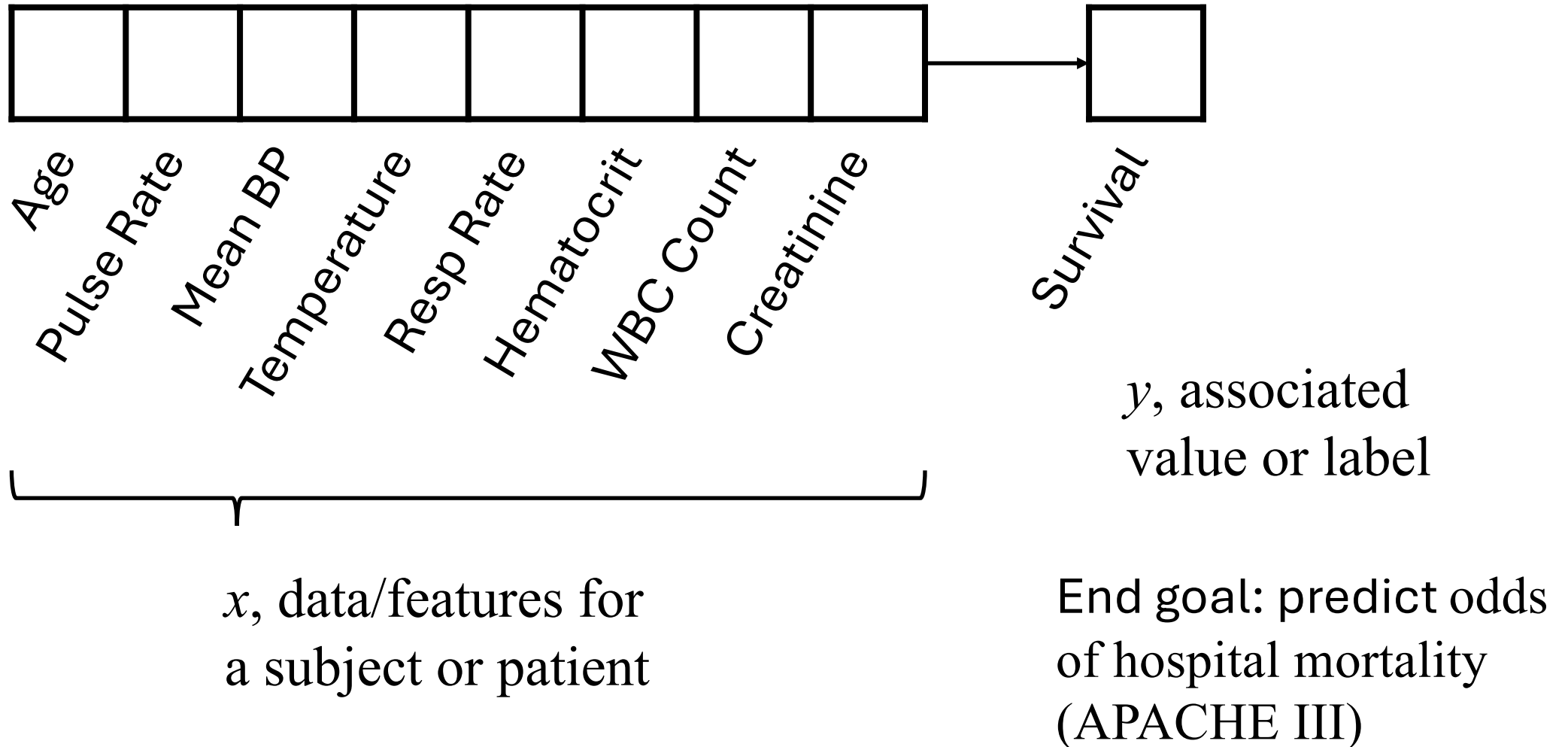
Overview of Predictive AI

The *machine* in machine learning



End goal: predict y from x

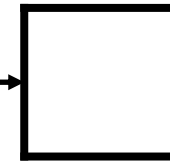
Simple models often work well for clinical data!



Computer vision: prediction via convolutional neural network



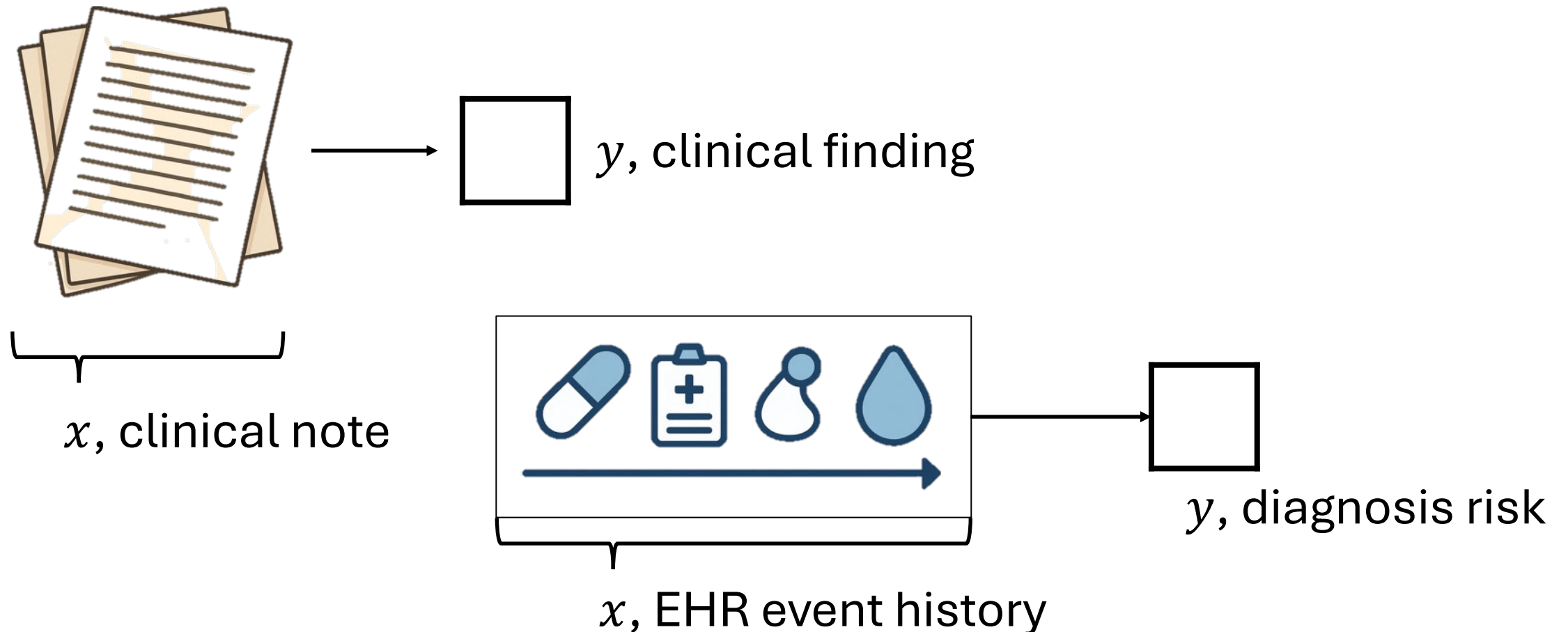
x , retinal image



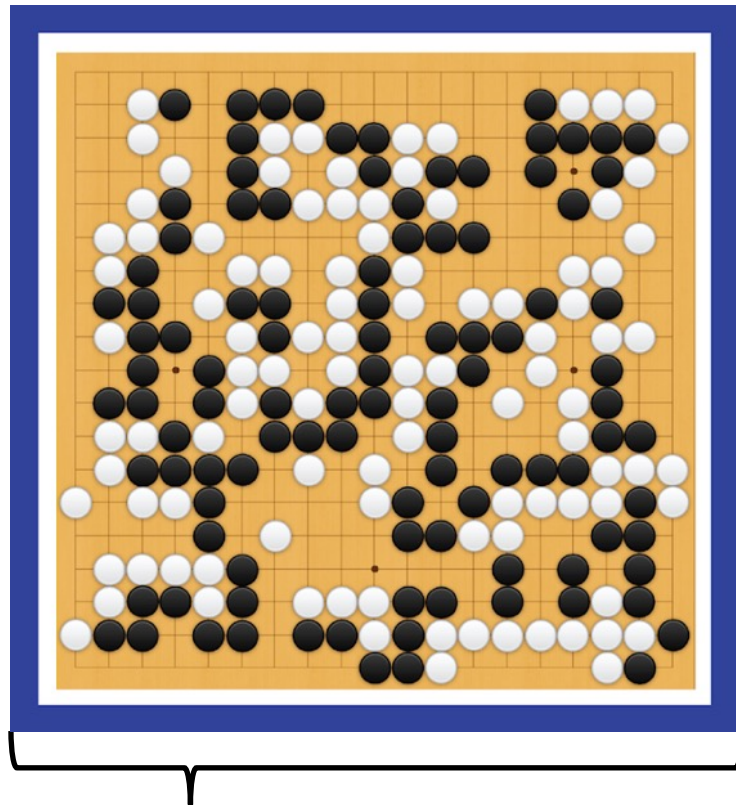
y , referable
diabetic retinopathy

End goal: predict y from x

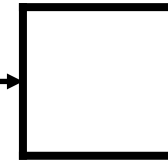
NLP and clinical sequences: prediction via stacked attention networks



Reinforcement Learning: state to action predictions via dense NN



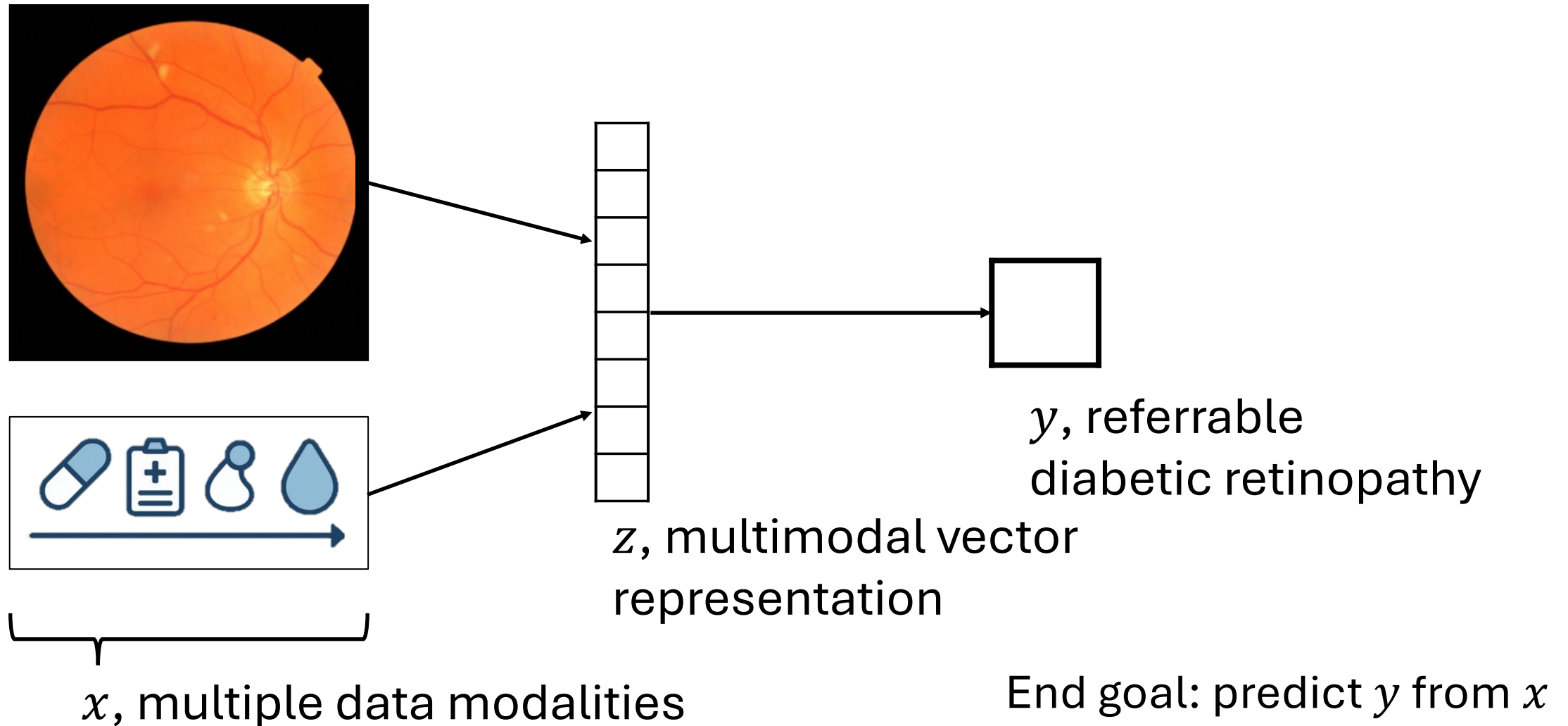
x , Go board state



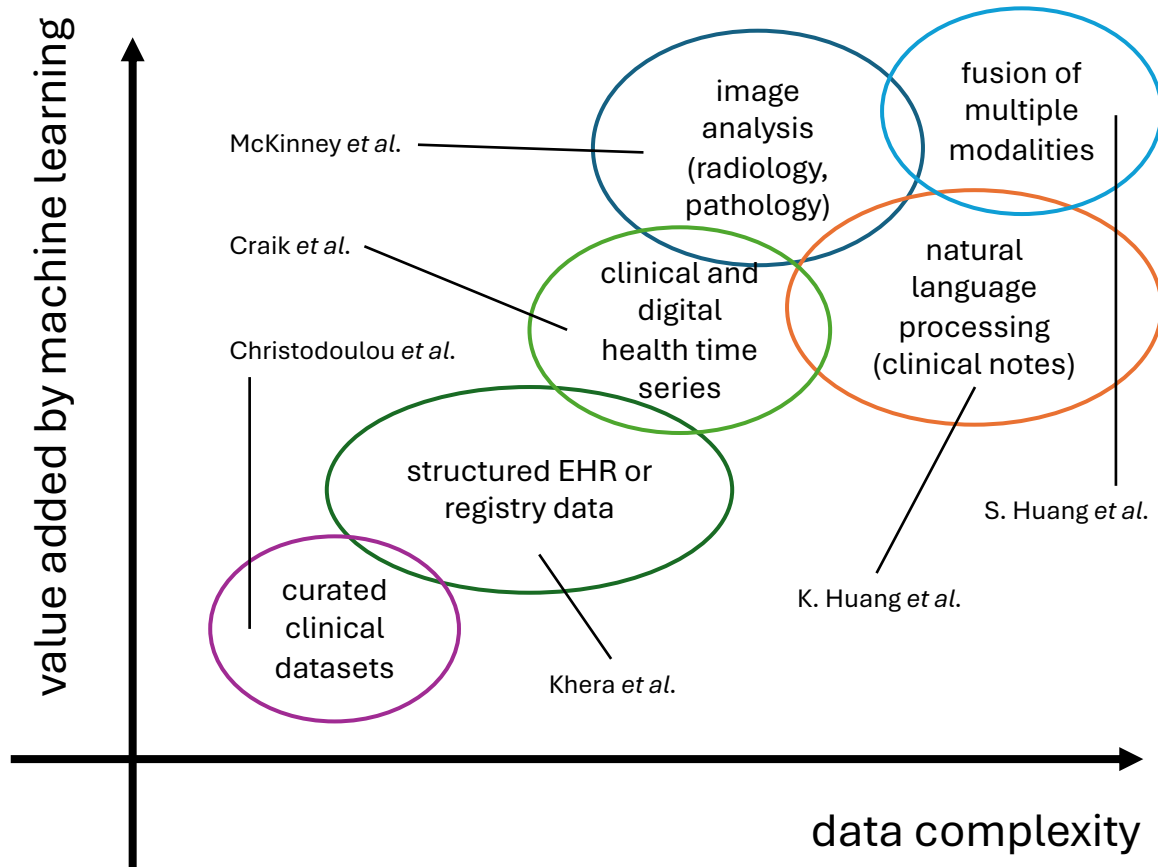
y , next move

End goal: predict y from x

Prediction from multimodal data: modality-specific encoders + multimodal fusion



As the complexity of the underlying data increases, so too does the probable value added by machine learning



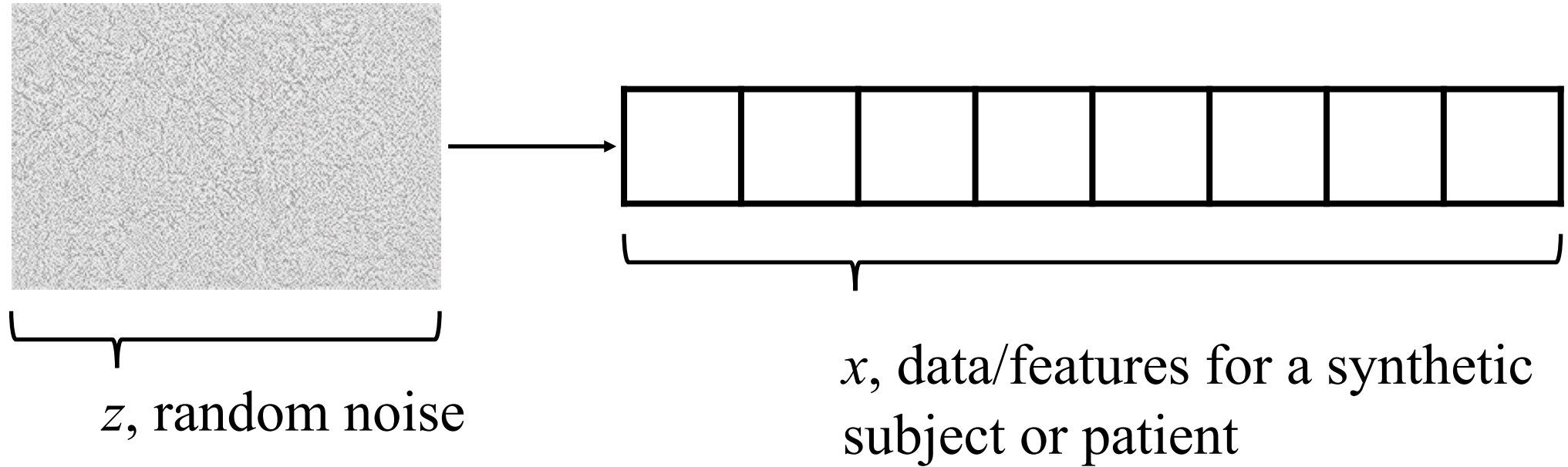
Simple models based on clinical variables remain highly relevant

Use the simplest model that gets the job done

Engelhard MM, Navar AM, Pencina MJ. Incremental benefits of machine learning—when do we need a better mousetrap?. JAMA cardiology. 2021 Jun 1;6(6):621-3.

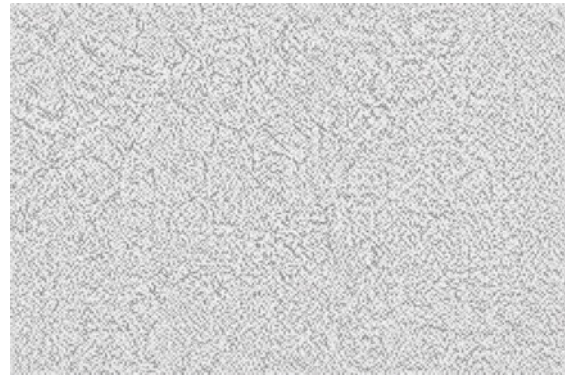
Overview of Generative AI

Generating tabular data



End goal: create structure from randomness

Image generation via GAN or diffusion



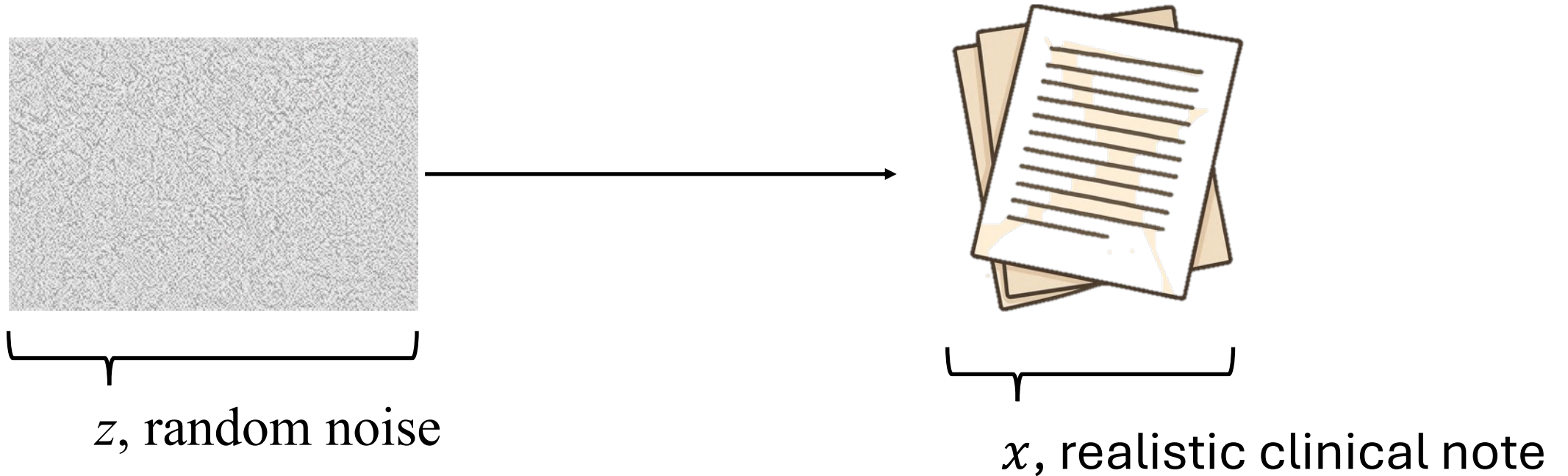
z , random noise



x , realistic retinal image

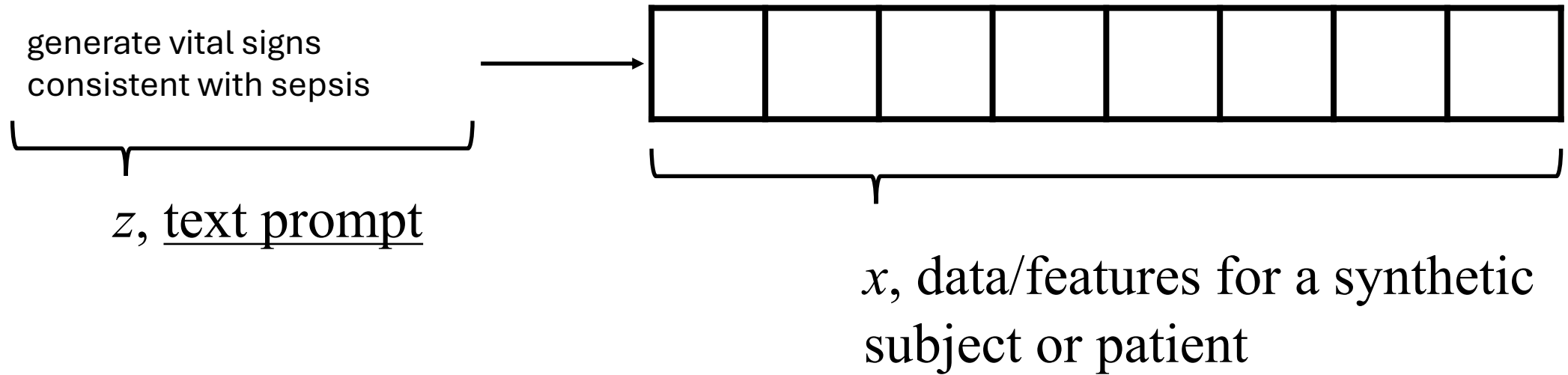
End goal: create structure from randomness

Text gen via Transformer decoder or diffusion



End goal: create structure from randomness

Prompt-based generation of tabular data



End goal: create structure
consistent with the user prompt

Prompt-based image generation

create a fundoscopic image
with exudates present

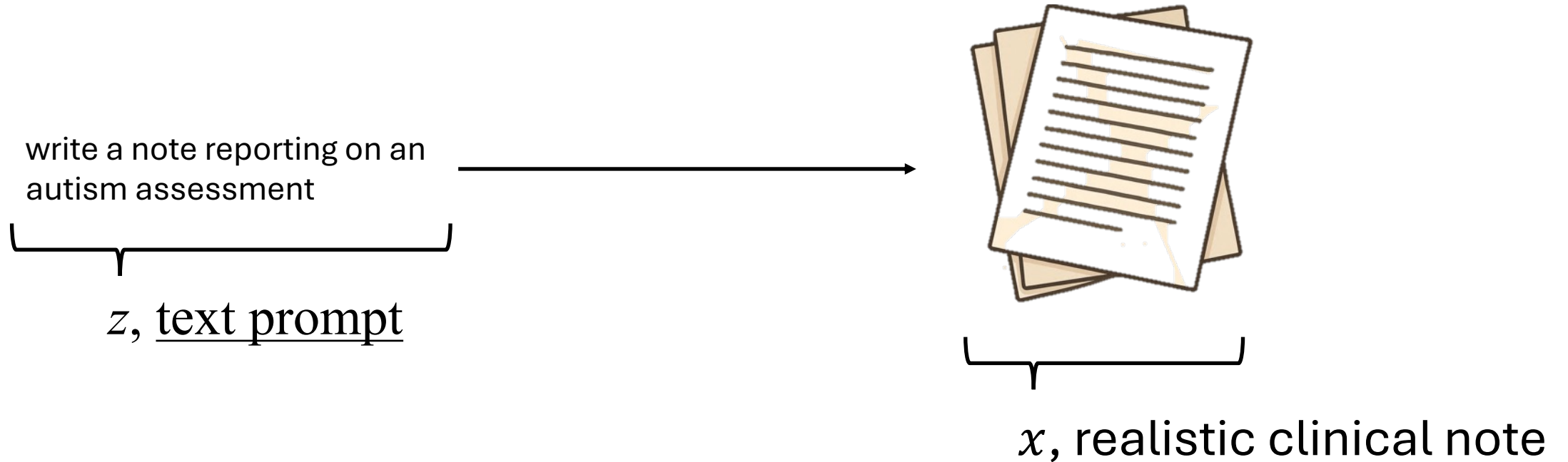
z , text prompt



x , realistic retinal
image

End goal: create structure
consistent with the user prompt

Prompt-based text generation



End goal: create structure
consistent with the user prompt

The prompt can be an image



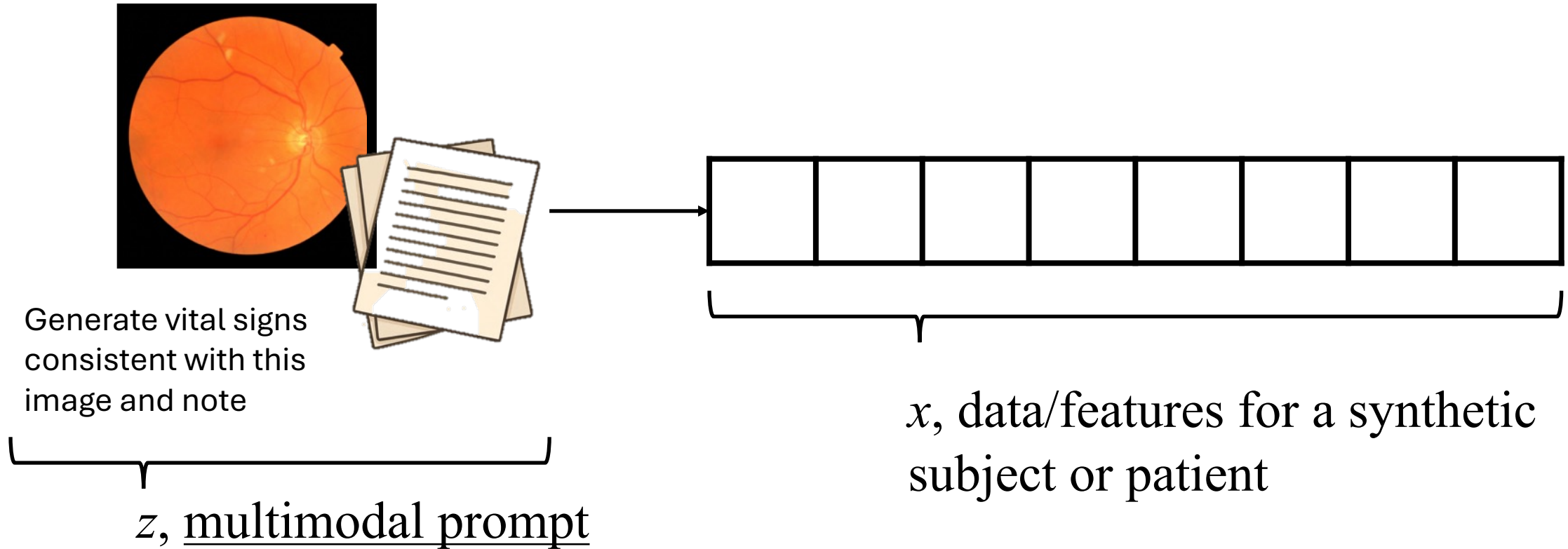
z , image prompt



x , realistic clinical note

End goal: create structure
consistent with the user prompt

The prompt can be complex...

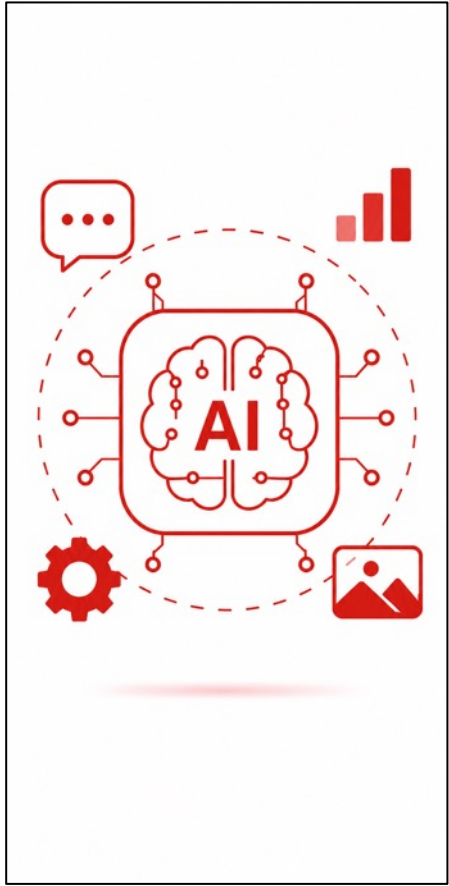


End goal: create structure
consistent with the user prompt

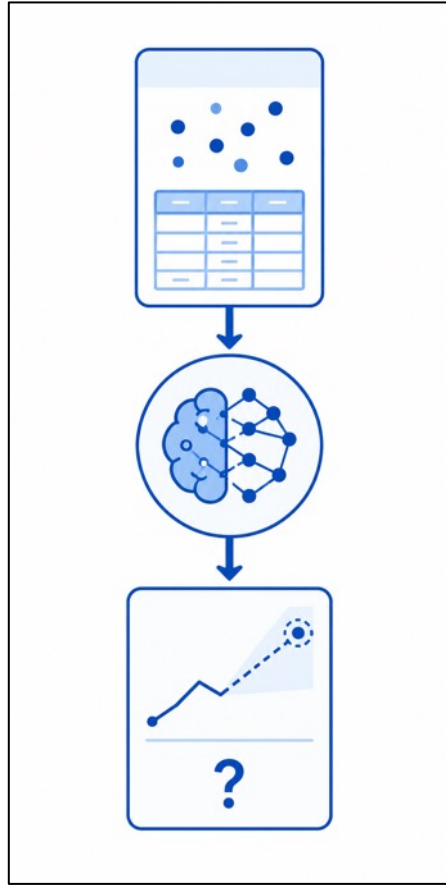
What we will cover in this class

A roadmap by course weekend

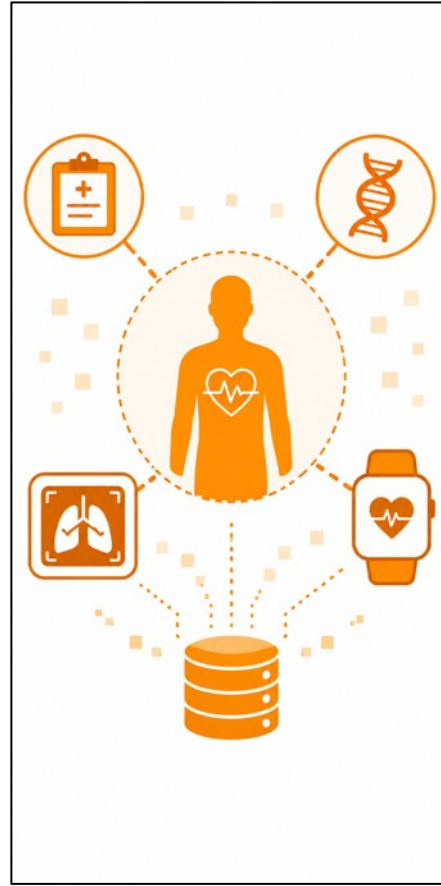
Content by course weekend



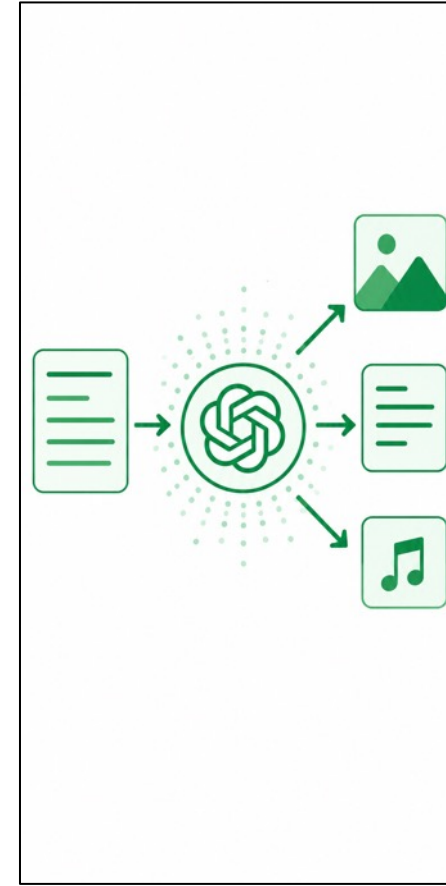
1: AI Overview



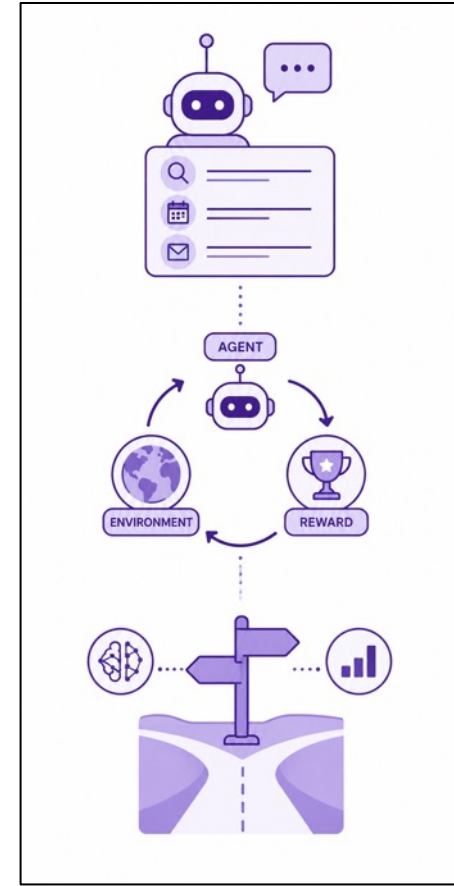
2: Predictive AI



3: Complex data

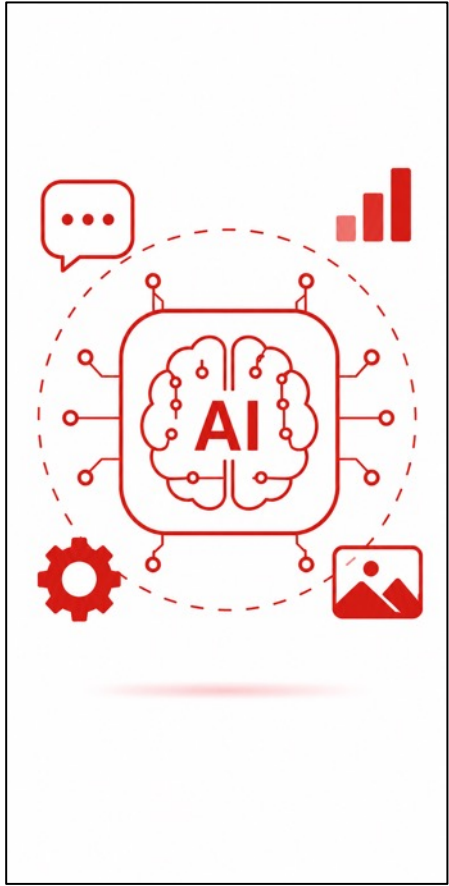


4: Generative AI

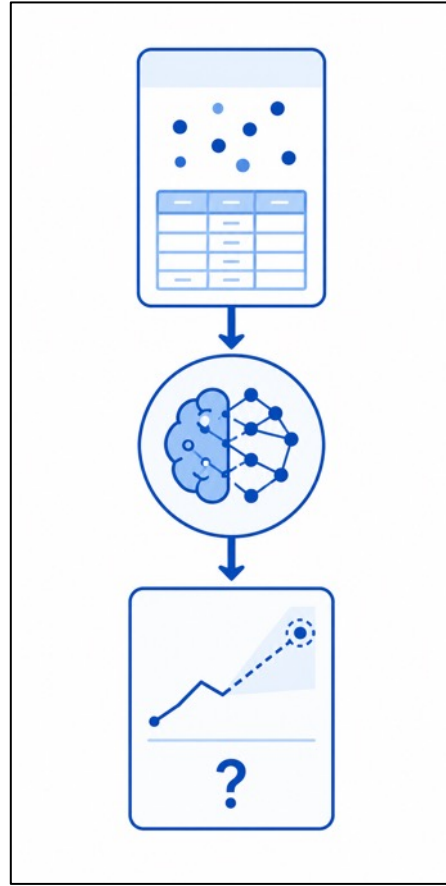


5: Emerging AI

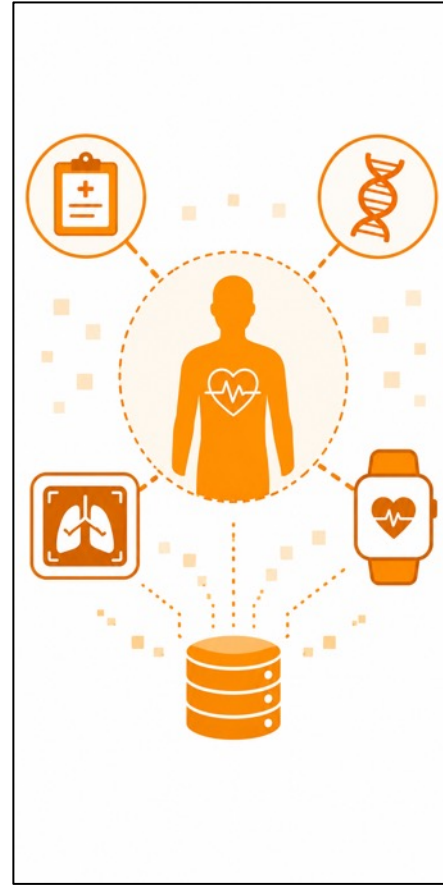
Content by course weekend



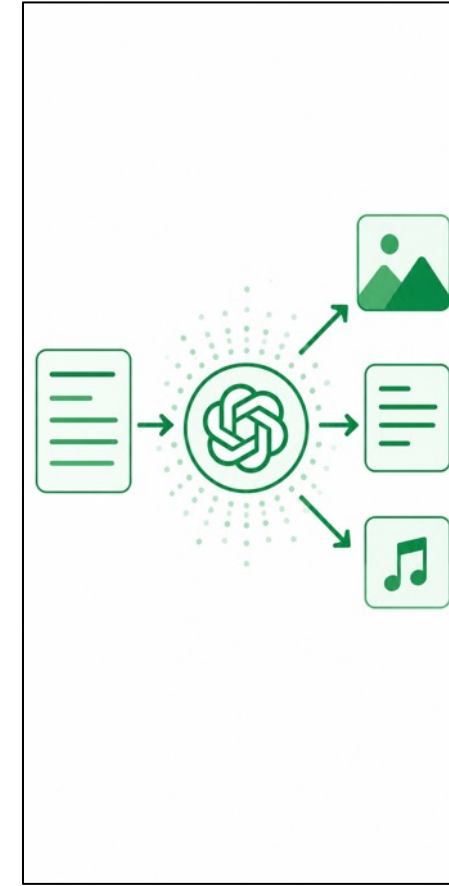
1: AI Overview



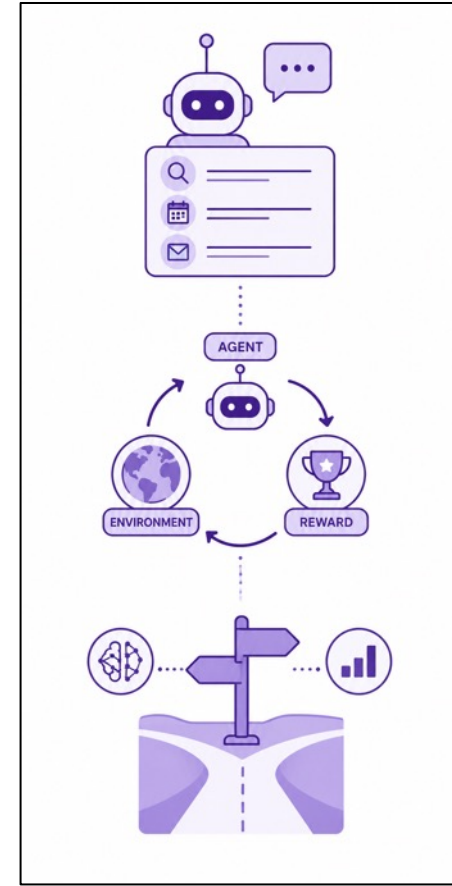
2: Predictive AI
+ how to evaluate



3: Complex data
+ how learning works



4: Generative AI
+ mitigating bias



5: Emerging AI
+ explaining AI

Activity 1

Predictive, Generative, or Both?

Activity 1: predictive, generative, or both?

Instructions: propose a predictive, generative, or hybrid system for one of the following tasks/problems, or one of your own:

- Improving management/handling of patient portal messages
- Managing and titrating hypertension medications
- Identifying social determinants of health from EHR data
- Improving cancer screening
- Drafting a discharge summary following a complex admission

Deliverables:

- In a few sentences, describe your proposed system at a high level
- Briefly explain why you believe generative, predictive, or hybrid is best

Learning Objective: consider tradeoffs and synergies between predictive and generative AI

Predictive, Generative, or Both?

Choosing the right tool or combination of tools

Pros and cons of predictive and generative AI



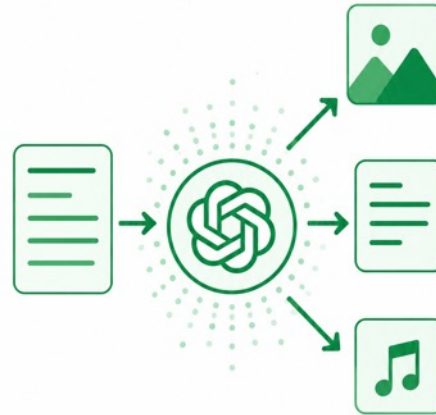
Predictive AI

Pros

- Pro 1

Cons

- Con 1



Generative AI

Pros

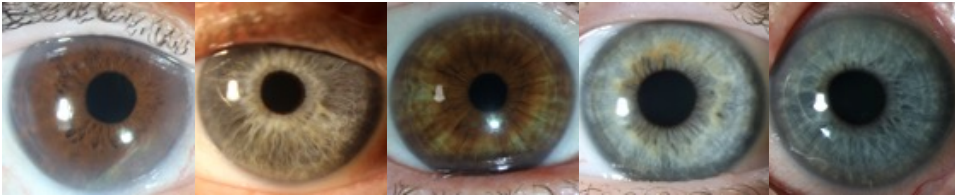
- Pro 1

Cons

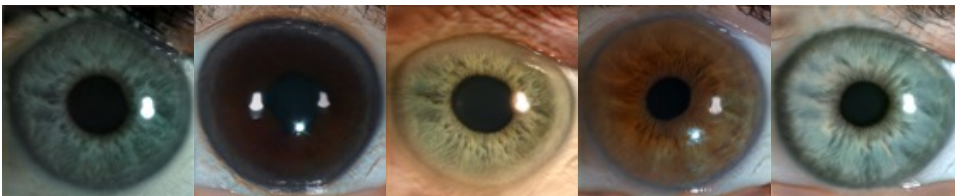
- Con 1

Often predictive and generative models are complementary.

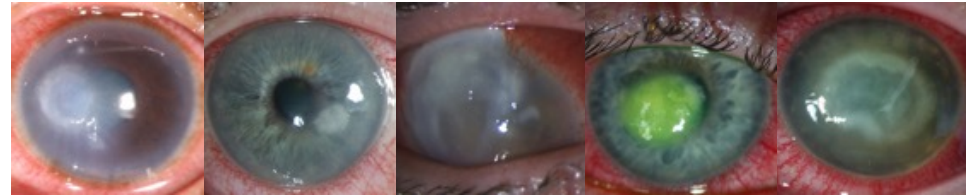
Real Healthy (n=57)



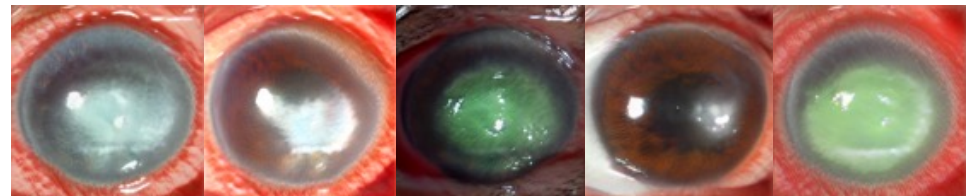
Synthetic Healthy



Real Diseased (n=27)



Synthetic Diseased



Microbial keratitis (MK) detection:

- Learn to generate synthetic data (generative)
- Train an MK detection model (predictive)

Often predictive and generative models are complementary.

Maternal health response system:

- Speech to text (predictive)
- Translation (predictive)
- Identification of key concepts and topics (predictive)
- Triage (predictive)
- Generation of template and/or complete responses in specific cases (generative)

It is much easier to evaluate the performance of predictive models, and in turn to know when and how much to trust them.



In some cases, however, generative models have superseded predictive models.

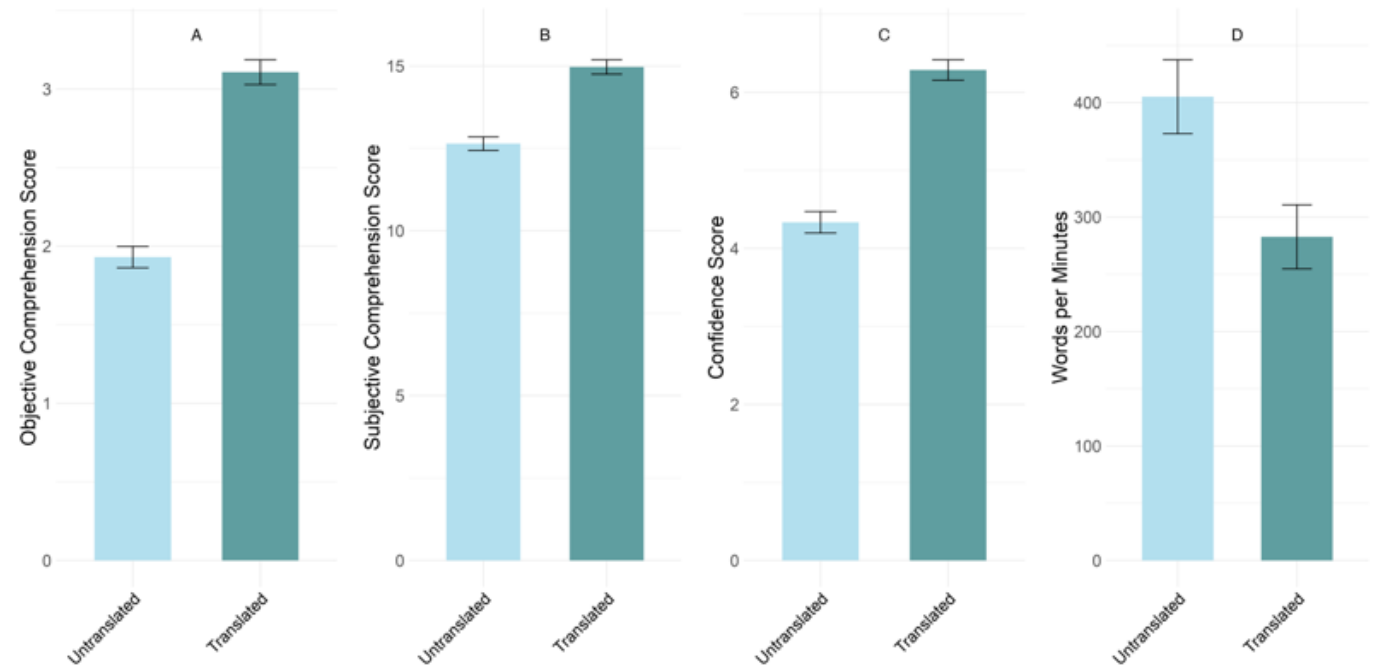
“Translation” of discharge notes into plain language using GPT4 substantially improves patients’ ability to comprehend them.



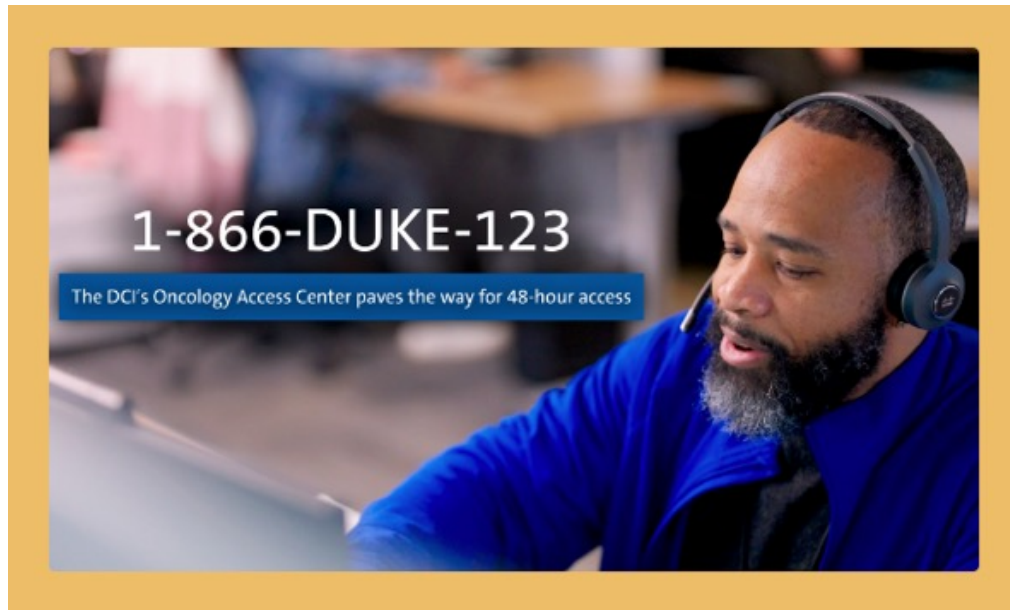
Anivarya Kumar



Isabella Wang



Extracting specific info from documents: predictive or generative?



Data needed to determine scheduling needs
ICD-10
Stage
Age
Date of Dx
Hormone Receptors and molecular biomarkers
History of Treatment (med, surg, rad)
Undiagnosed patients only: Review of symptoms

While a predictive approach might be more trustworthy in principle, generative models can perform many tasks “zero-shot”, i.e. without task-specific training.



Pros and cons of predictive and generative AI



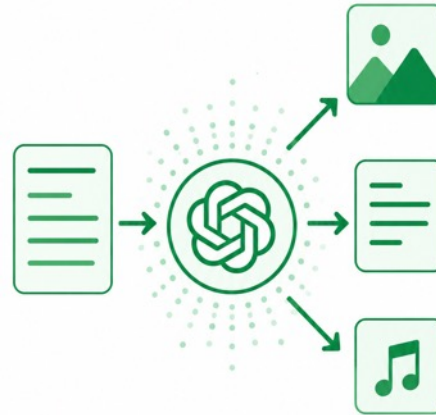
Predictive AI

Pros

- Pro 1

Cons

- Con 1



Generative AI

Pros

- Pro 1

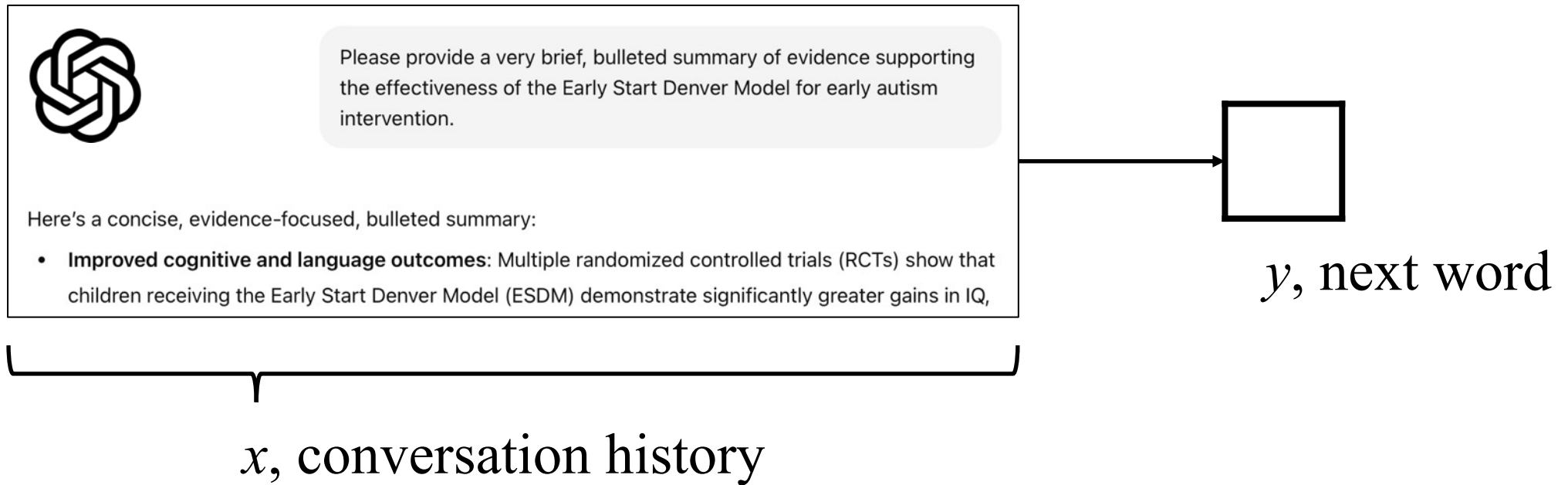
Cons

- Con 1

A blurry line

Between predictive and generative AI

Is this generative or predictive?



“next-token prediction”

Are all tasks special cases of text generation?

Model: GPT-4

ME

Please read the following echocardiogram report and tell me whether left ventricular hypertrophy was detected. Answer yes or no only. 

Echo reading:

MILD LV SYSTOLIC DYSFUNCTION (See above) WITH MILD LVH
NORMAL RIGHT VENTRICULAR SYSTOLIC FUNCTION MODERATE
VALVULAR REGURGITATION (See above) NO VALVULAR STENOSIS
NO PERICARDIAL EFFUSION SEVERE BIATRIAL ENLARGEMENT AT
LEAST MILD PULMONARY HTN ELEVATED RA PRESSURES NO
PRIOR STUDY FOR COMPARISON



Yes



Are all tasks special cases of text generation?

ME



Please read the following echocardiogram report again and tell me whether left ventricular hypertrophy was detected. This time, please give me your confidence level as a percentage ranging from 0% (i.e. you are certain that it was NOT detected) to 100% (i.e. you are certain that it WAS detected). Answer with the percentage only.

Echo reading:

MILD LV SYSTOLIC DYSFUNCTION (See above) WITH MILD LVH
NORMAL RIGHT VENTRICULAR SYSTOLIC FUNCTION MODERATE
VALVULAR REGURGITATION (See above) NO VALVULAR STENOSIS
NO PERICARDIAL EFFUSION SEVERE BIATRIAL ENLARGEMENT AT
LEAST MILD PULMONARY HTN ELEVATED RA PRESSURES NO
PRIOR STUDY FOR COMPARISON



100%



Are all tasks special cases of text generation?

ME

OK. Now, please read the following text message and tell me whether you believe the sender was angry when sending it. Please give me your confidence level as a percentage ranging from 0% (i.e. you are certain that they were NOT angry) to 100% (i.e. you are certain that they WERE angry). Answer with the percentage only.



> Hey, you didn't show up today. What's the deal?



60%



Activity 2 (time permitting)

Anatomy of logistic regression

Course Logistics

- In class: lectures, group activities, and discussion
- Group assignments
- Final project
- AI/LLM policy

What you will need

- Willingness to learn and engage
- Patience!
- A ChatGPT or Claude subscription...